

Daily Review



1 A transistor in a circuit acts as a...

Switch

2 A relay lets a small current switch...

A separate, bigger
circuit

3 A diode across a relay coil protects
the...

Transistor

4 It gets dark. An LDR's resistance...

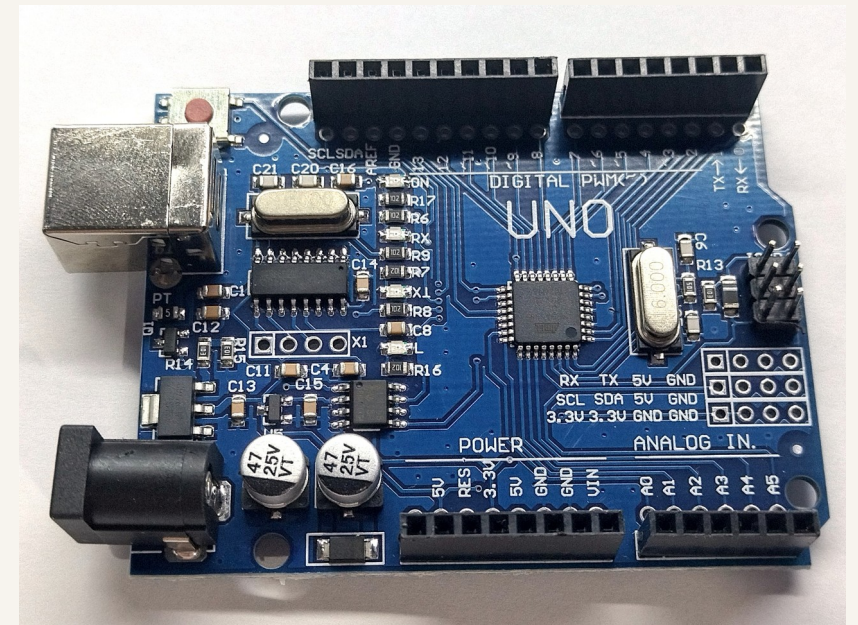
Rises

5 Three parts of a system diagram?

Input, process, output

Microcontrollers

One chip runs a washing machine, a bus door and a packing line.



National 5 Engineering Science

How does a washing machine know when to spin?



A washing machine

PART 1

What a Microcontroller Is

One chip, one program, a whole machine.

By the end of this part...



- 1 I can describe what a microcontroller is
- 2 I can give commercial and industrial examples
- 3 I can give advantages and disadvantages against hard-wired control

What Is a Microcontroller?



Microcontroller

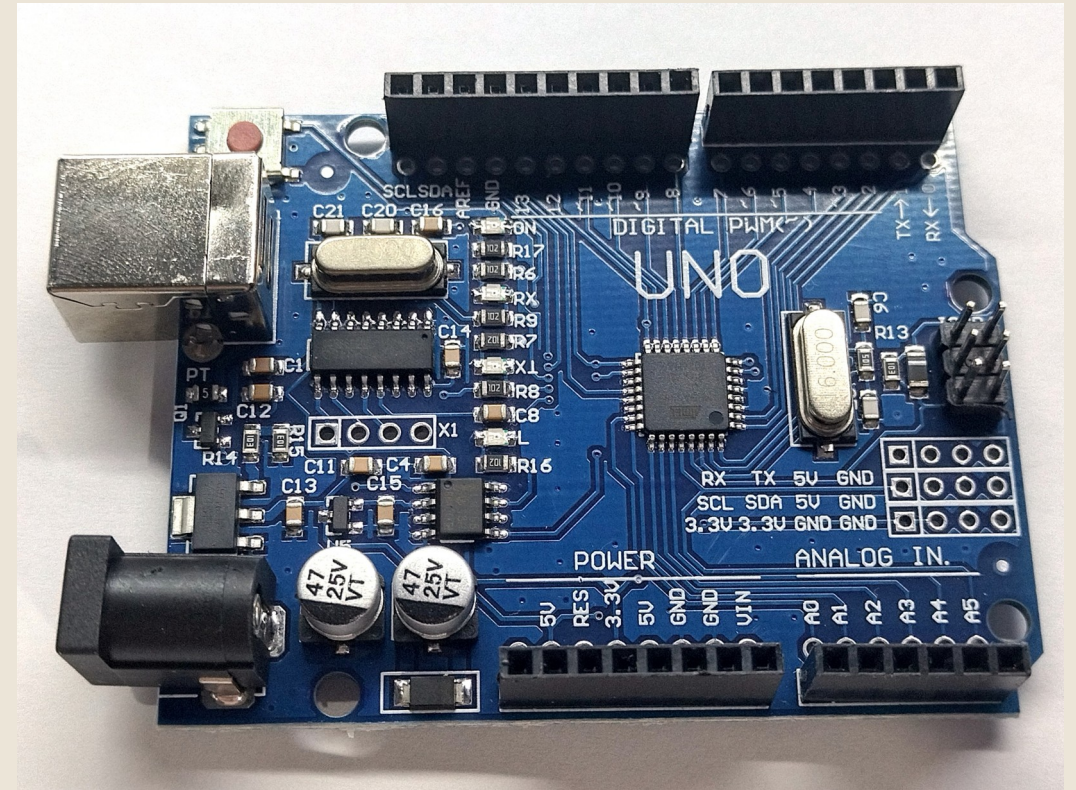
a small, programmable chip
that controls a system

It runs

a program stored in its
memory

It reads

inputs, and switches outputs



Input, Process, Output

Input

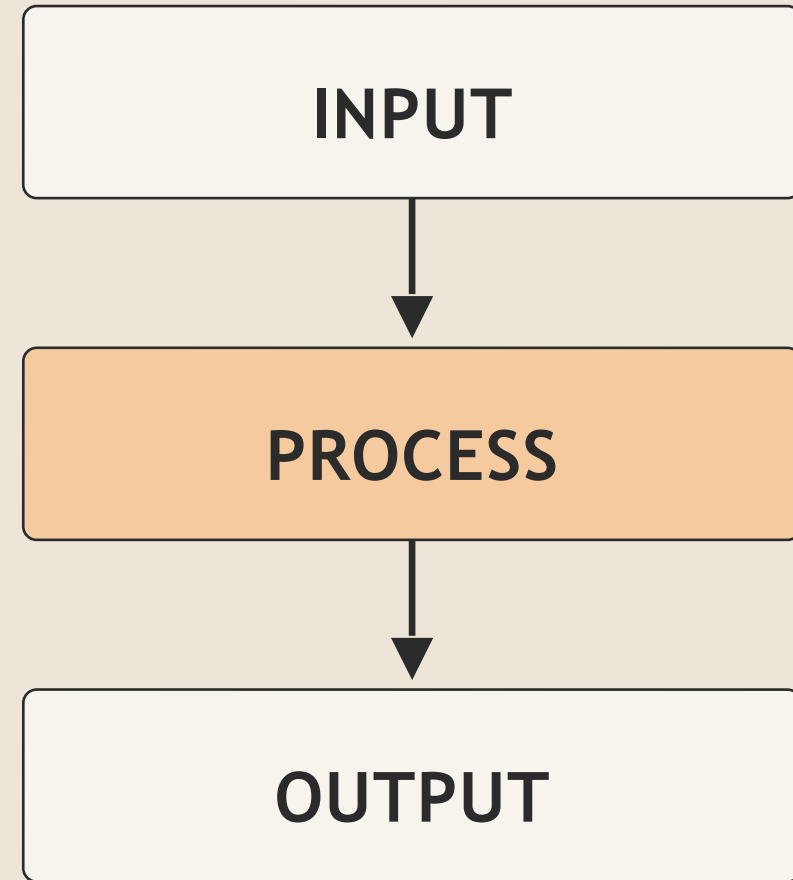
a switch or a sensor

Process

the microcontroller and its program

Output

a lamp, buzzer or motor



Where You Meet Them



Commercial
ticket machine



Industrial
packing line



Transport
bus doors



Home
smart lighting

Which Sector?



1	An alarm panel in a shop	Commercial
2	A lift in an office block	Industrial
3	Anti-lock brakes on a car	Transport
4	Central heating control	Home
5	A packaging machine	Industrial

Hard-Wired Control

Hard-wired

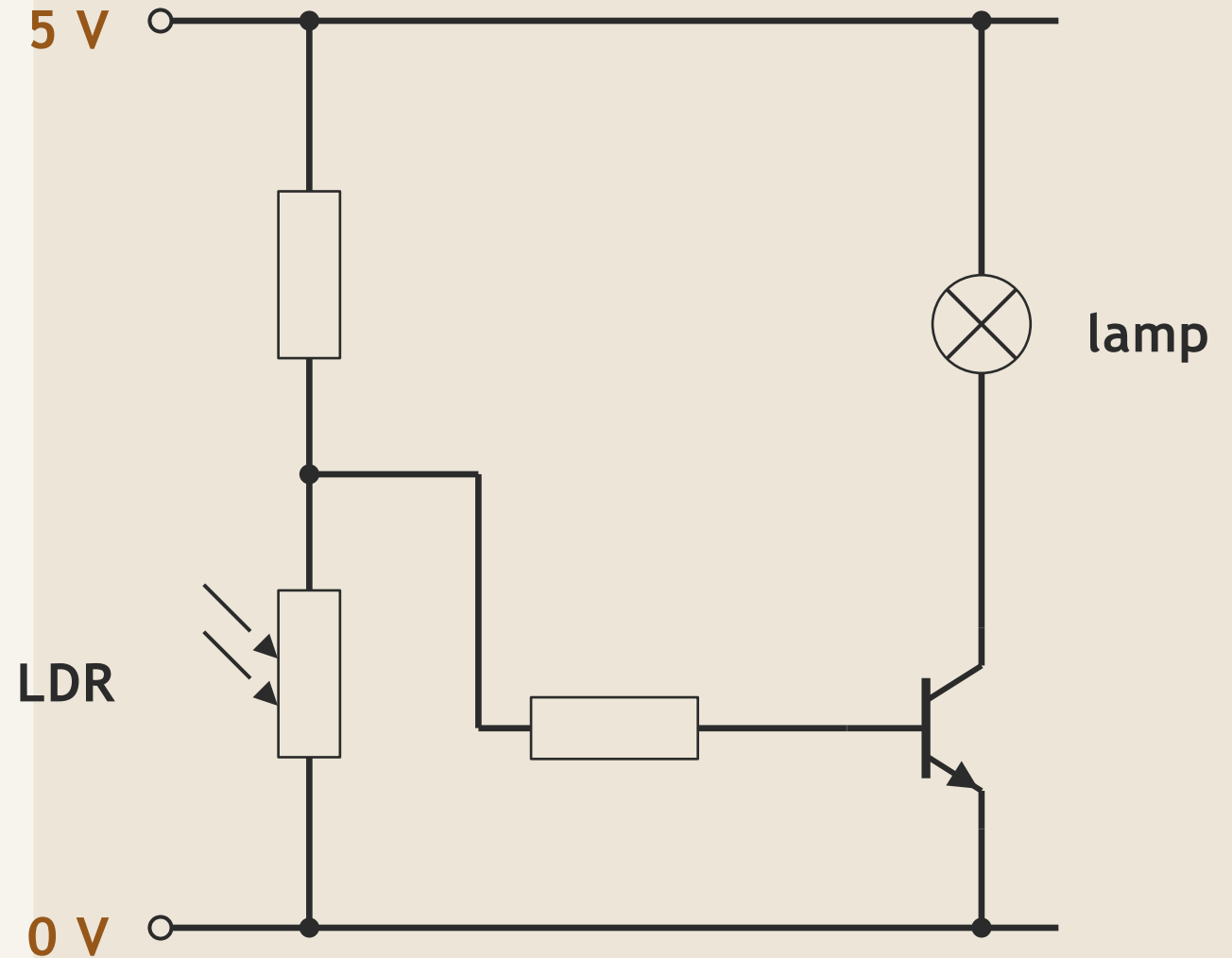
fixed components and wires

The job is

in the wiring

Change the job

new parts, new wiring



Programmable Control

Programmable

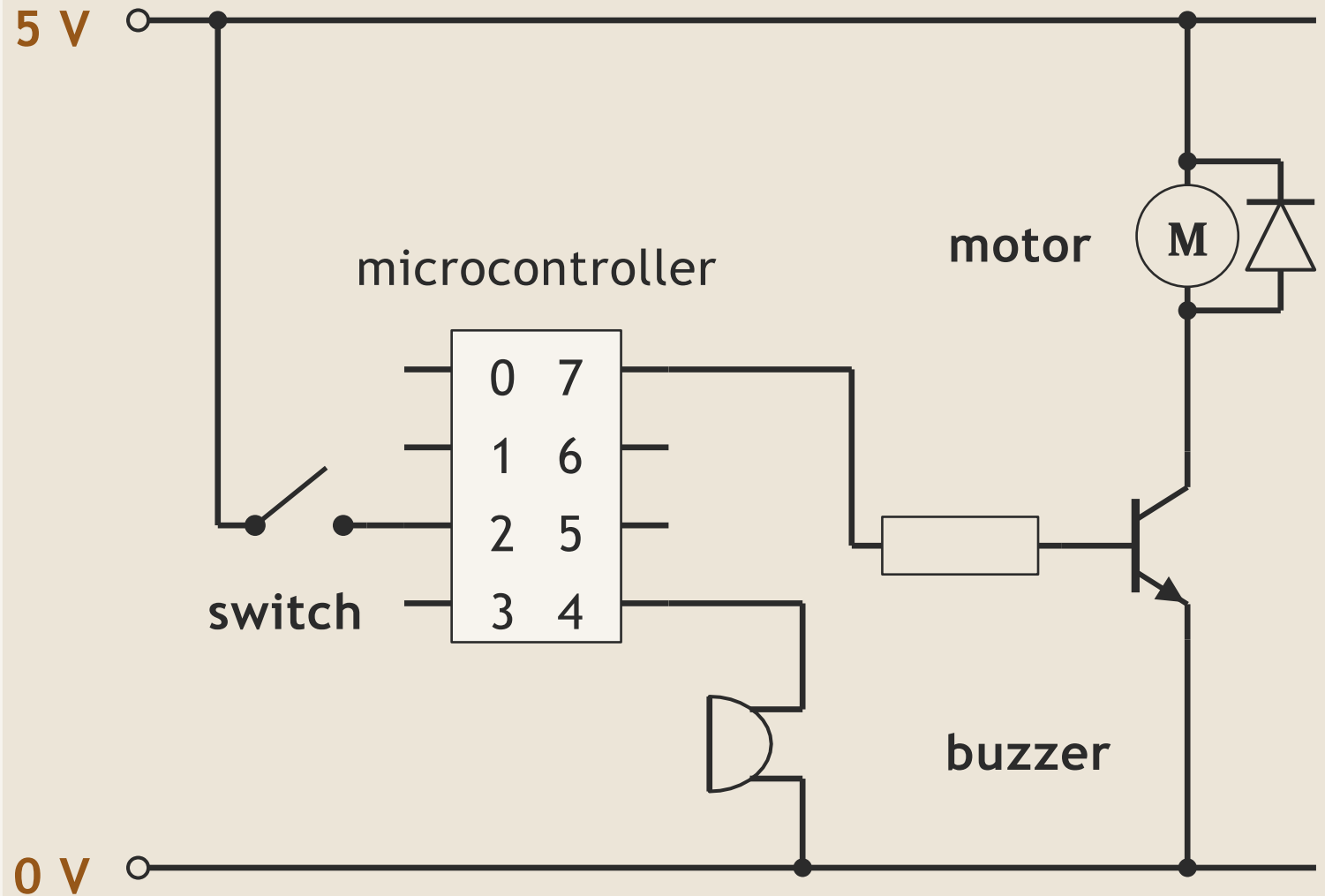
a microcontroller
and a program

The job is

in the program

Change the job

change the
program



Advantages



Reprogrammed

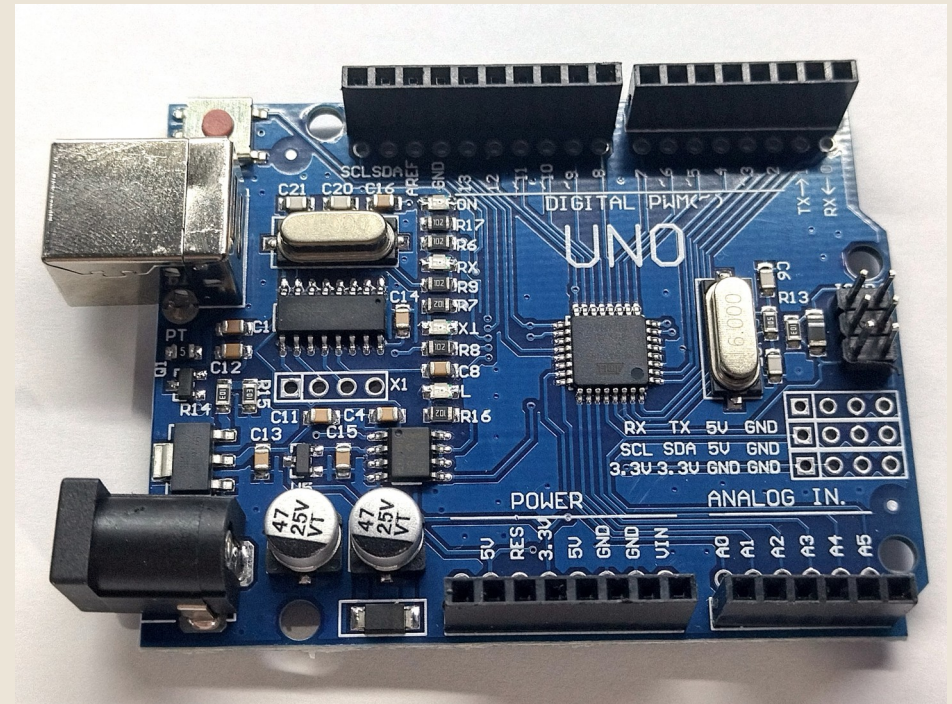
change the program, not the wiring

Re-used

one board, many products

Fewer components

so more reliable, faster to assemble



Disadvantages



Needs a programmer

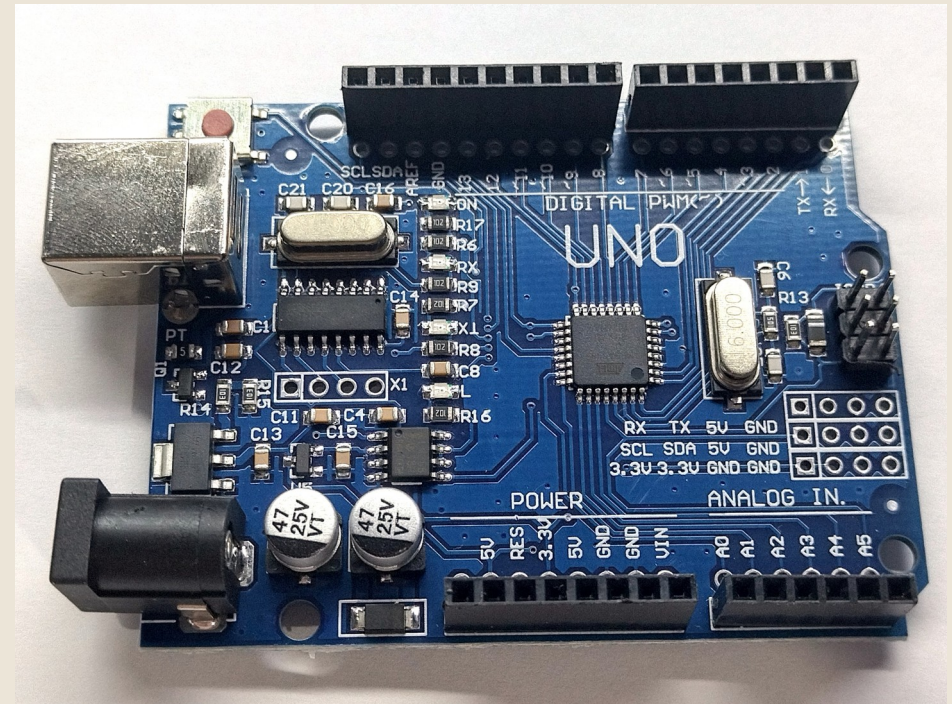
someone must write and test the program

Needs a power supply

the chip needs one to run

Harder to follow

you cannot see the job by looking at the parts



Watch Out: “Easier to Change”

THE MISTAKE

- It is easier to change.
- It is cheaper.
- It is programmable.

THE CORRECT ANSWER

- It can be reprogrammed, with no rewiring.
- It needs fewer components, so it costs less.
- One board can be re-used for a new job.

Model Answer: Describe Two Advantages (2 marks)



Describe two advantages of a microcontroller over a hard-wired circuit for a bus's doors.

1 (1 mark)

The door times can be reprogrammed with no rewiring.

2 (1 mark)

Fewer components, so the doors are more reliable.



Model Answer: Explain Why (2 marks)



Explain why a microcontroller is used in a ticket machine.

Cause (1 mark)

The program can be updated

Effect (1 mark)

so new fares are added with no change to the circuit.



Your Turn: Same Form



1

Smart lighting: describe one advantage.

Reprogrammed — so no
rewiring

2

A packing line: explain why (cause + effect).

Fewer parts — so more
reliable

3

A pupil writes 'it is cheaper'. Marks?

0 on its own: say why it
costs less

Spot the Mistake



Describe two advantages of a microcontroller for a car-park barrier.

A pupil wrote

1 It is cheaper. 2 It is programmable.

Your job

How many marks? Why? Rewrite it.

TASK 1: Microcontrollers in Use



Booklet page 3, TASK 1

For each product, say why a microcontroller suits it.
Use the advantage words.

5 min



TASK 1: Answers



1 Microwave oven

Timed sequence; reads the door

2 Traffic lights

Timings changed by program, not rewiring

3 Robot arm in a factory

Runs motors in order; reprogrammed

4 Smart heating thermostat

Reads a sensor; settings set by program

Past Papers: Advantages



Past papers

Year	Question
2021	Q7(b)
2022	Q14(c)
2025	Q8(c)
2026	Q13(e)



By topic

Checkpoint



- 1 What is a microcontroller?
- 2 Give one disadvantage.
- 3 Why use one board for 3 alarm models?

A chip that runs a stored program

Someone must write the program

Re-usable — so just a new program

Plenary: Part 1



1 A microcontroller runs a stored...

Program

2 Give an industrial example.

Packing line, lift, robot
arm

3 Two advantages, MI words?

Reprogrammed; fewer
components

Plenary: Part 1



4 One disadvantage?

Needs a programmer
(or power supply)

5 What scores 0 on its own?

Easier to change,
cheaper, smaller

Look back — by the end of this part...



- 1 I can describe what a microcontroller is
- 2 I can give commercial and industrial examples
- 3 I can give advantages and disadvantages against hard-wired control

Daily Review



1 One advantage of a microcontroller?

It can be
reprogrammed

2 'Easier to change' alone scores...

0 marks

3 A transistor switches on at a base voltage of...

About 0.7 V

4 The base resistor limits the...

Base current

5 A packing line: commercial or industrial?

Industrial

PART 2

Wiring to the Pins

The lamp is too dim. What do you add?

By the end of this part...



- 1 I can wire an input switch to a pin
- 2 I can wire an output to a pin
- 3 I can wire a transistor driver to a pin
- 4 I can complete a wiring diagram from an I/O table

The Exam's Microcontroller

8 pins

each an input or
an output

The I/O table

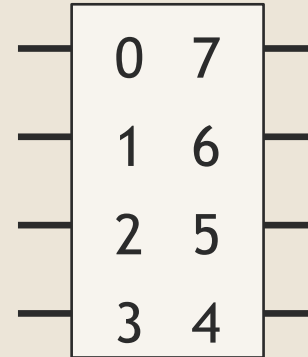
says which

In our room

in: 2, 3 out:
4 to 7

5 V ○

microcontroller



0 V ○

Wiring an Input



Switch

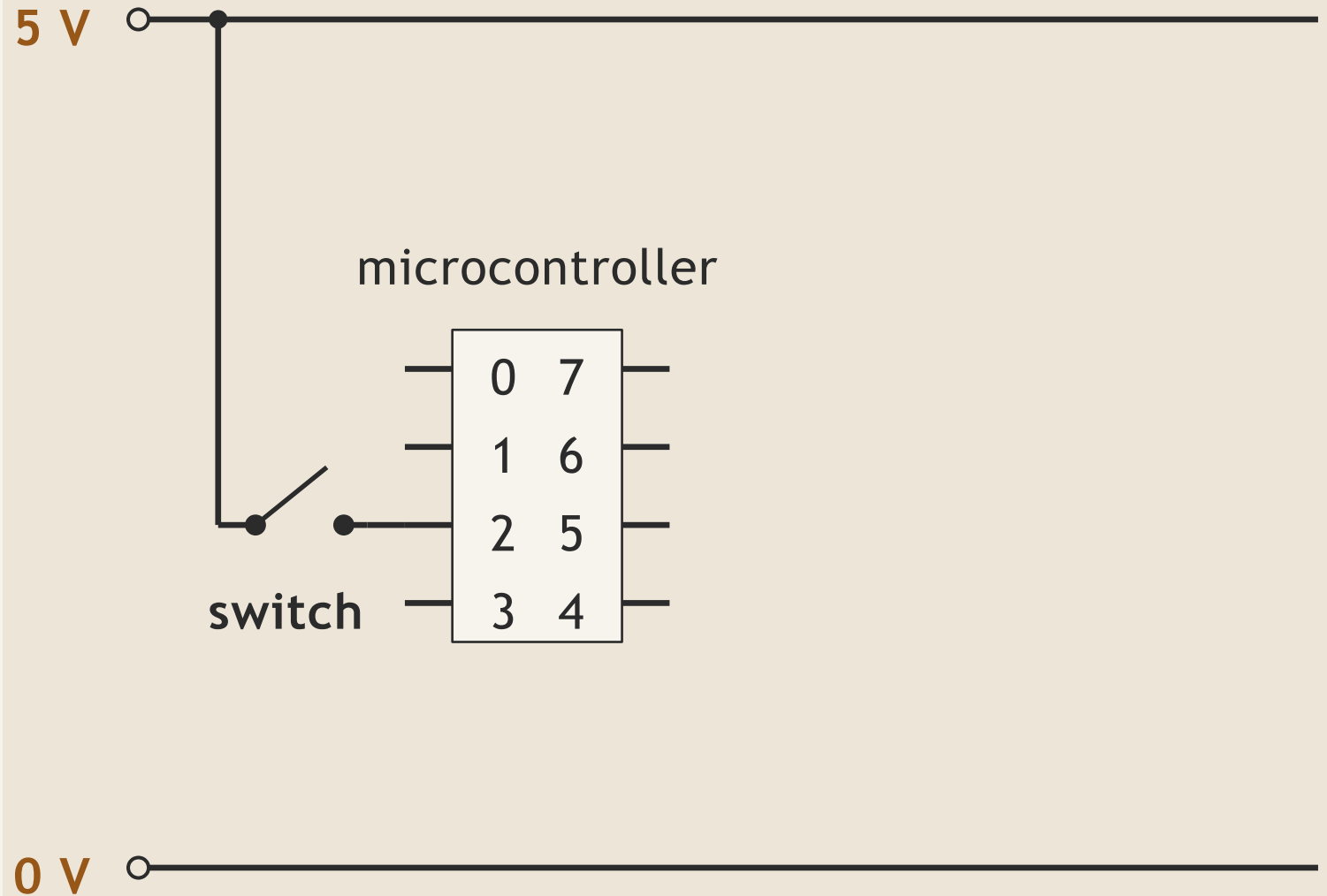
from 5 V to the
pin

Pressed

the pin gets 5 V

The program

reads the input as
on



Wiring an Output



Output

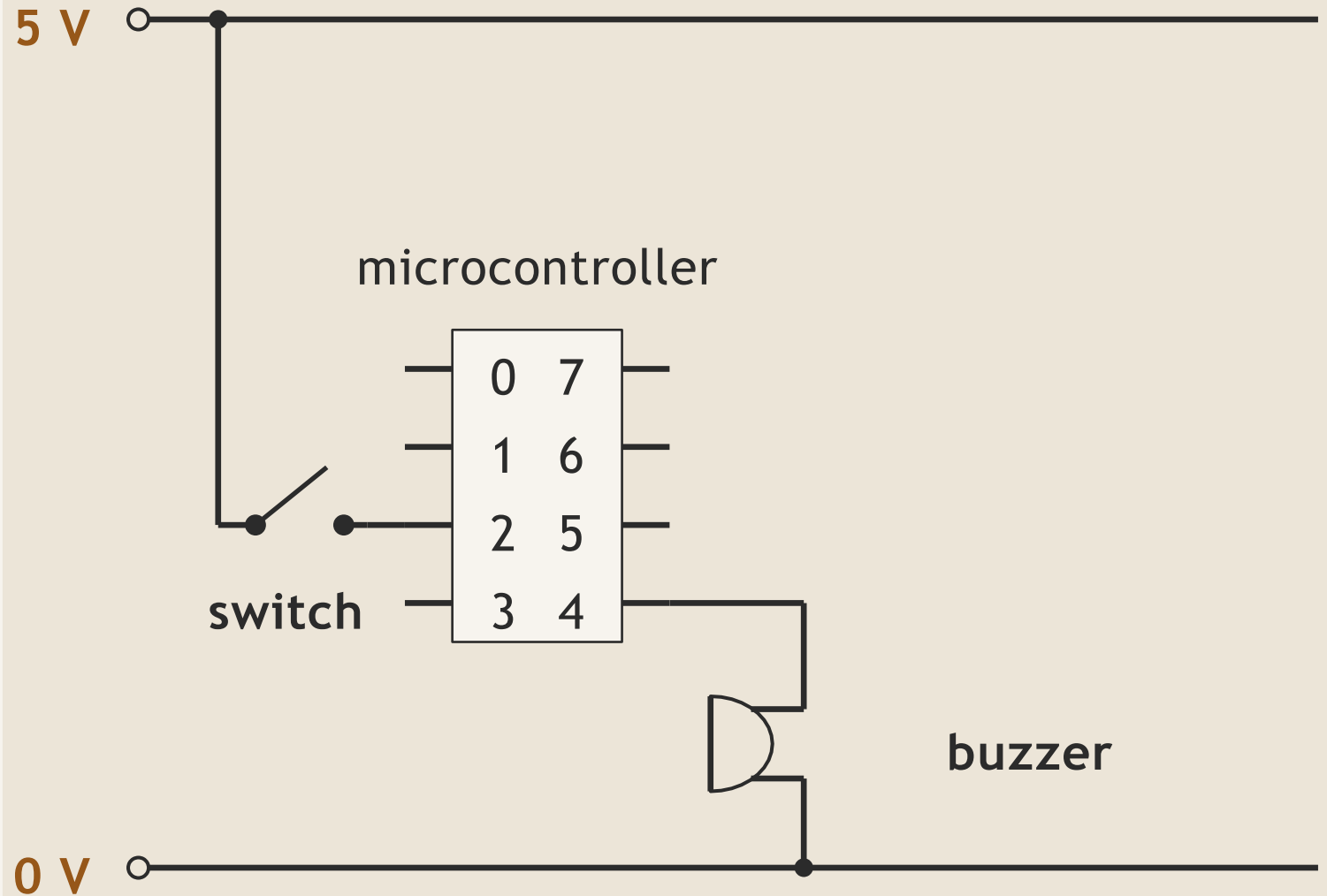
from the pin to
0 V

Switched on

5 V on the pin

Current flows

the buzzer sounds



A Motor Needs a Transistor

A pin gives

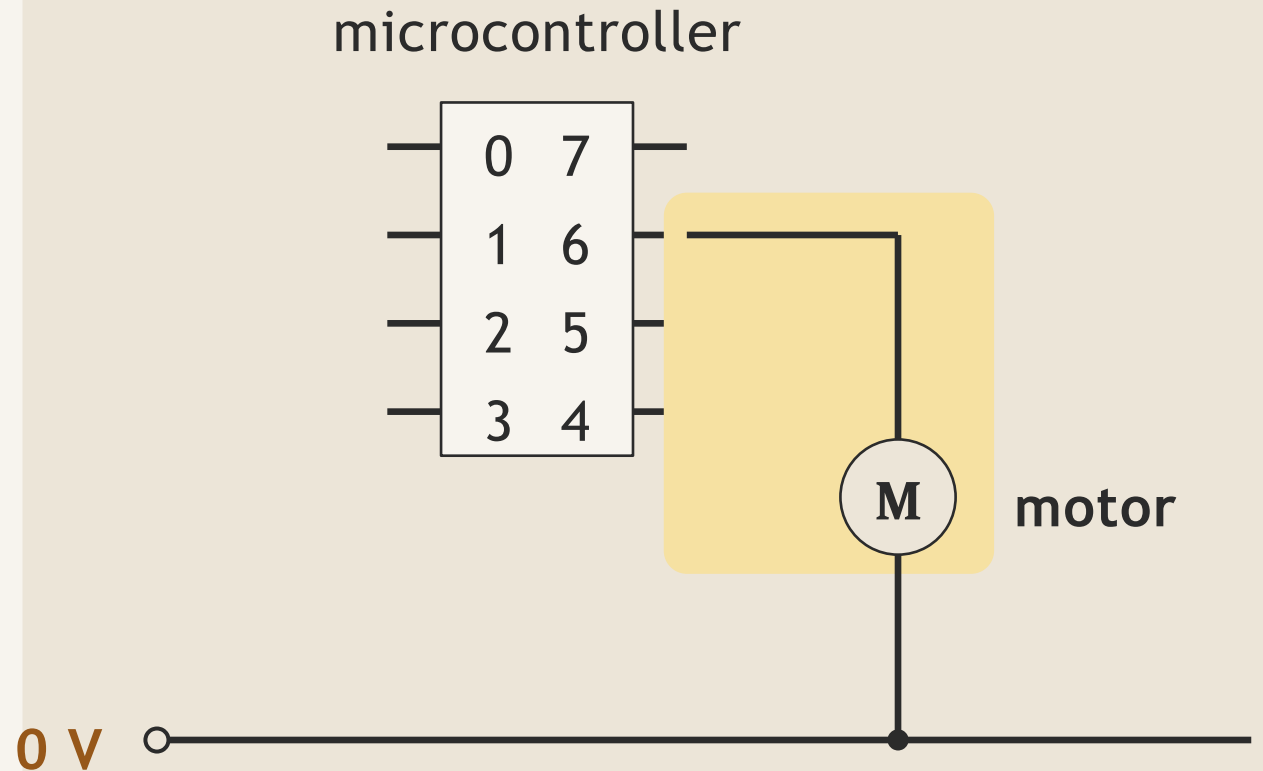
a small current, about
20 mA

A motor needs

much more

Add a transistor

“driver” alone scores 0



The Transistor Driver



Base lead

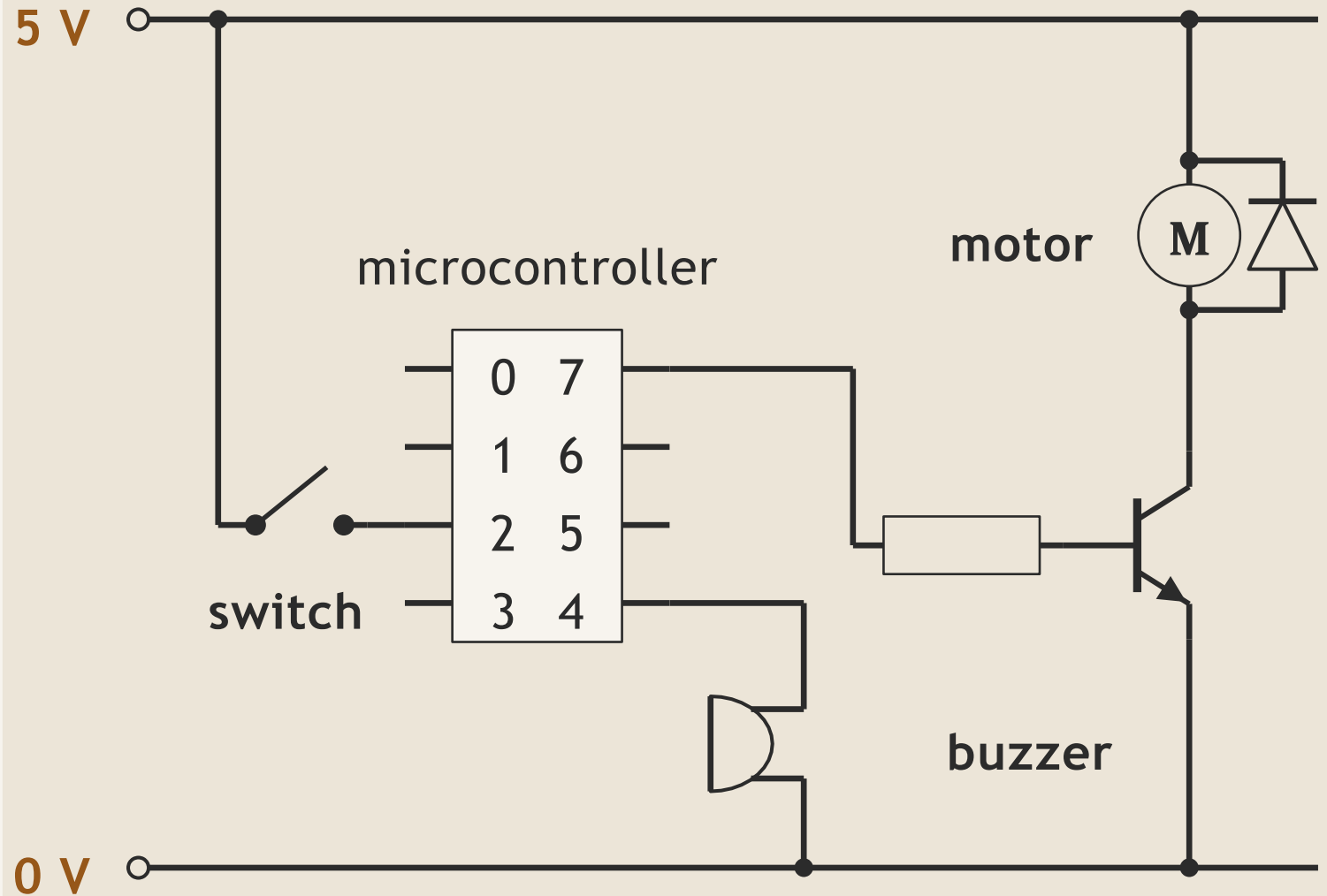
pin, resistor, base

Collector

motor, diode
across it

Emitter

to 0 V



Model Answer: Complete the Wiring (3 marks)



Input, pin 2

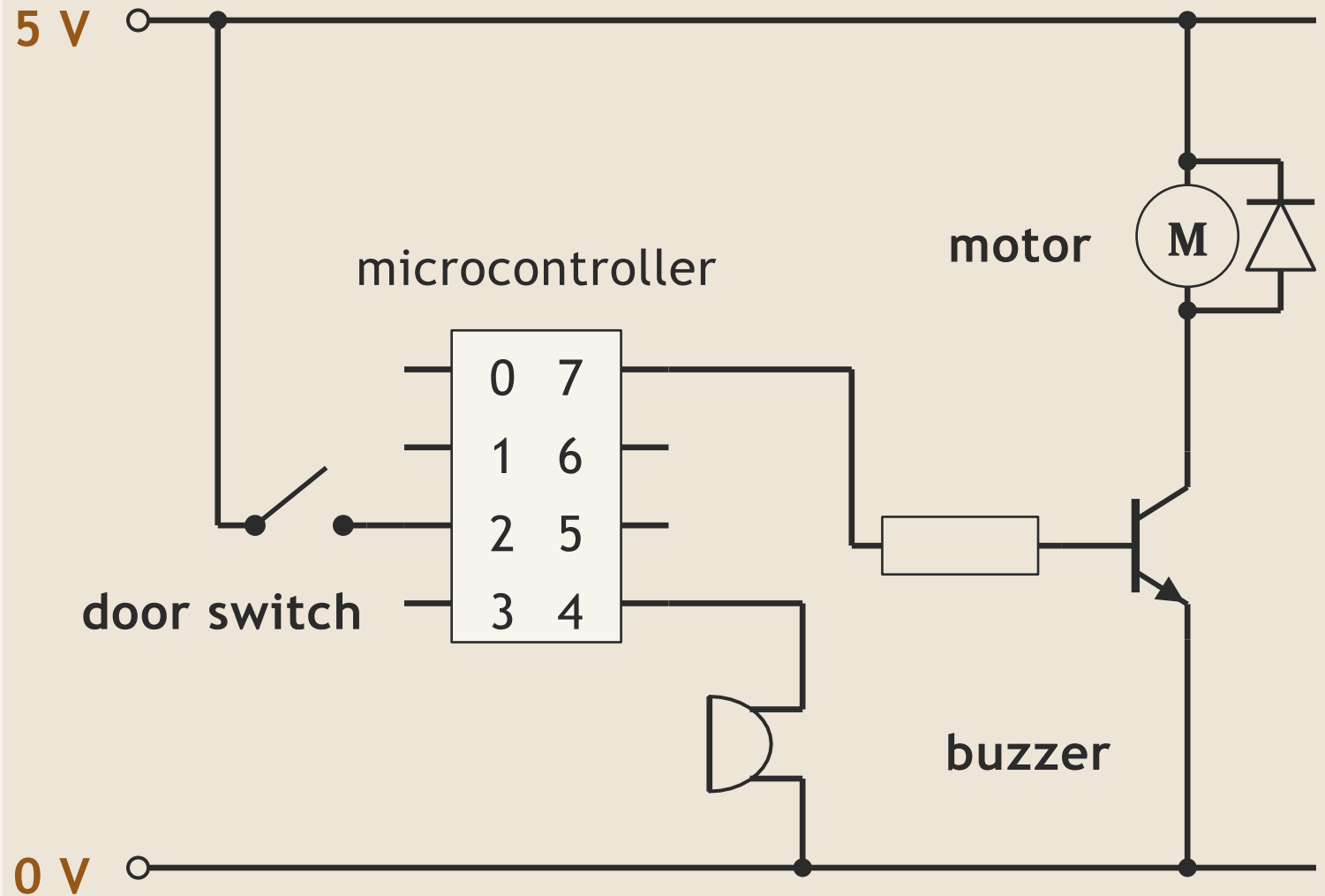
door switch

Output, pin 4

buzzer

Output, pin 7

motor, via a
transistor



The Same Circuit in the Simulator

Same I/O table

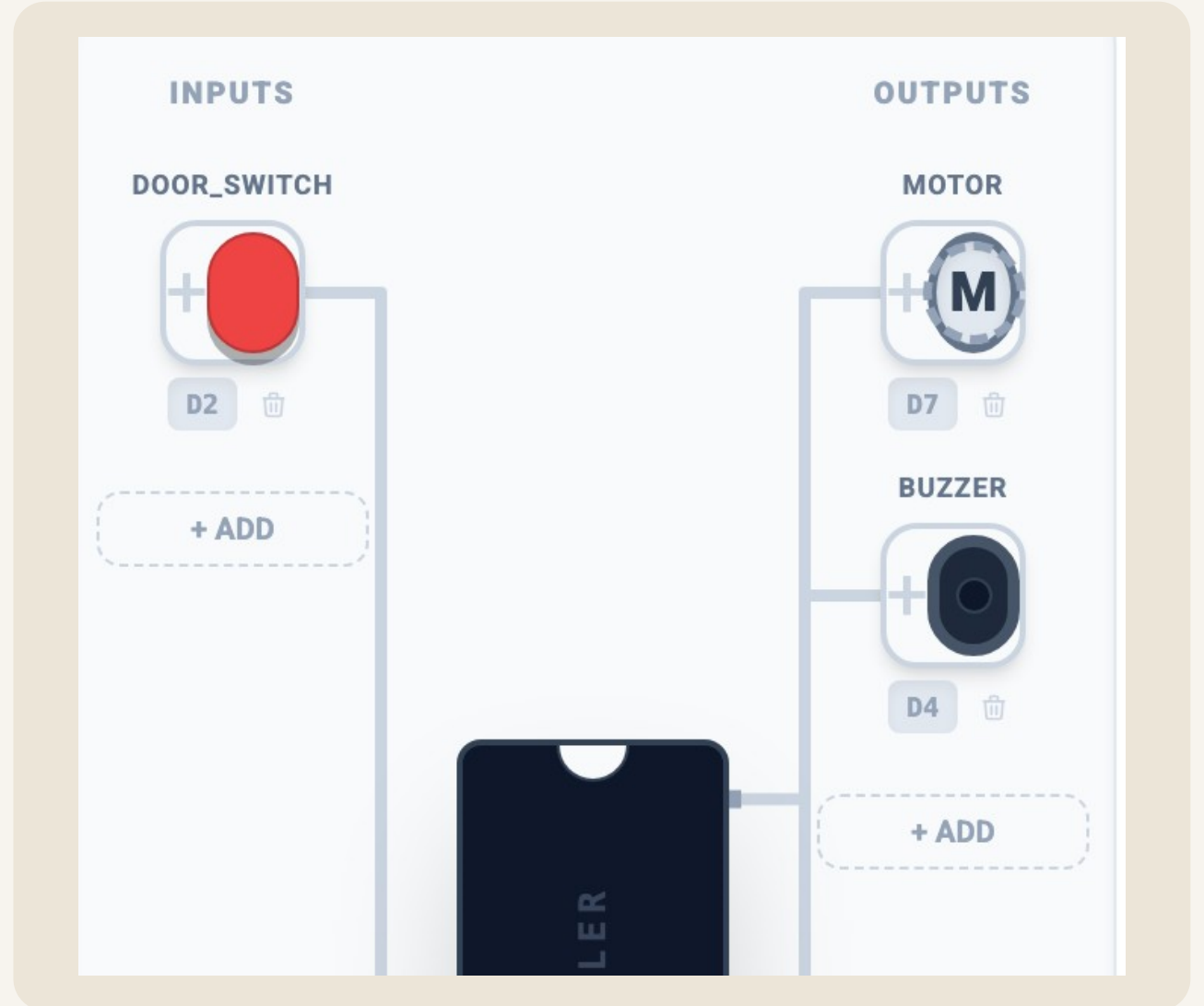
D2 in; D7 and D4 out

Each part

on its own pin

The sim hides

the transistor and the rails



Your Turn: Complete the Wiring



Input, pin 3

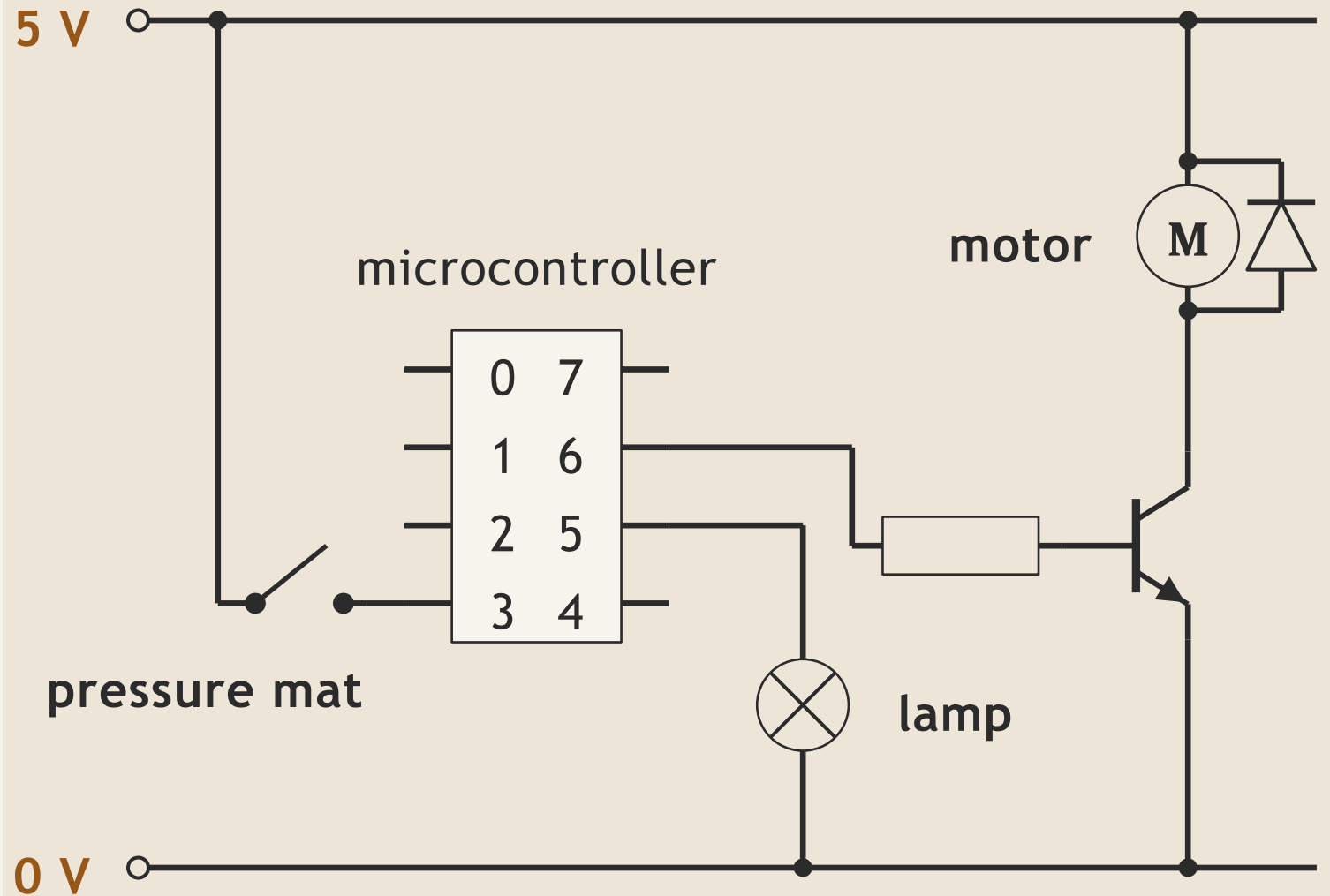
pressure mat

Output, pin 5

lamp

Output, pin 6

motor, via a
transistor



Spot the Mistake



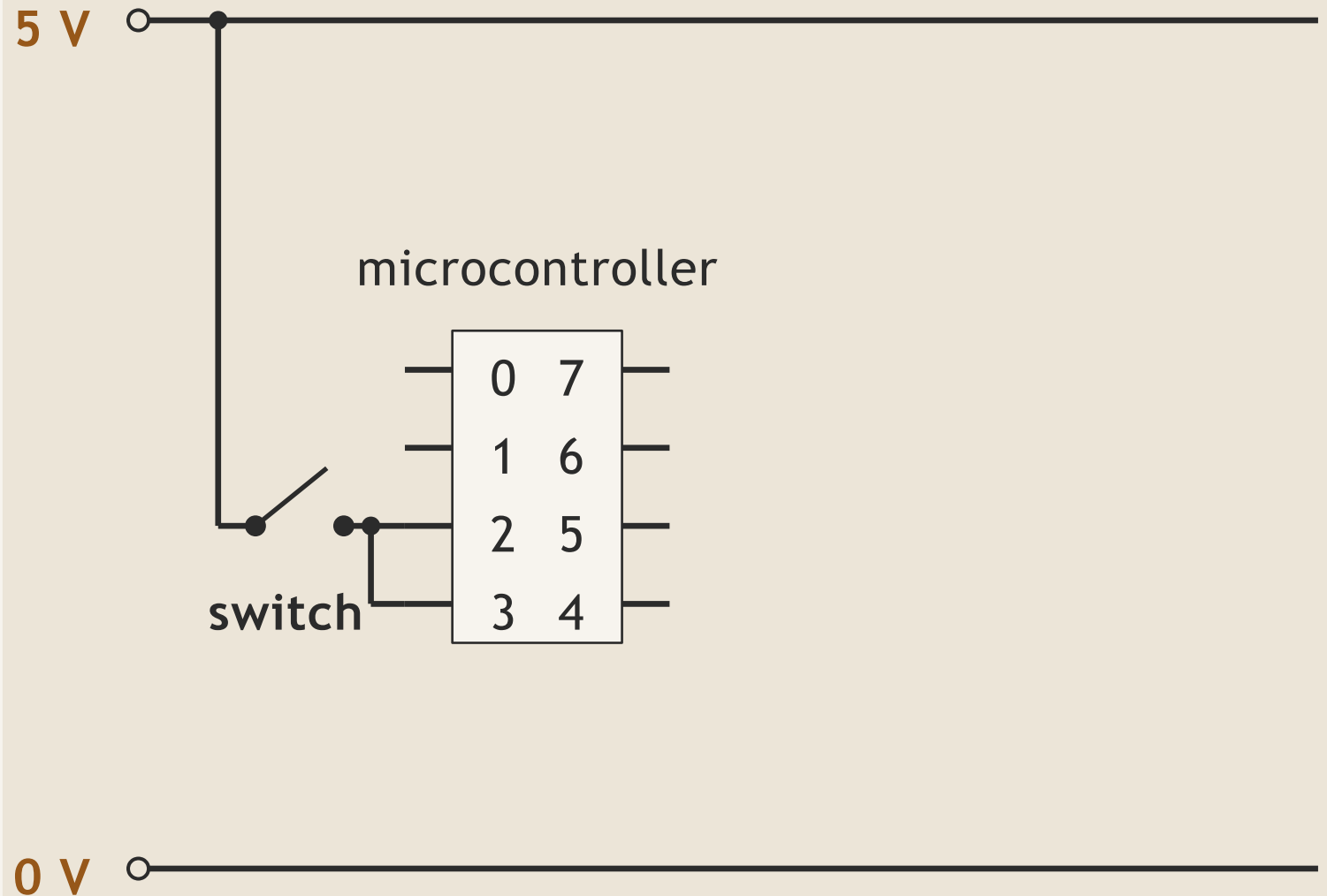
I/O table

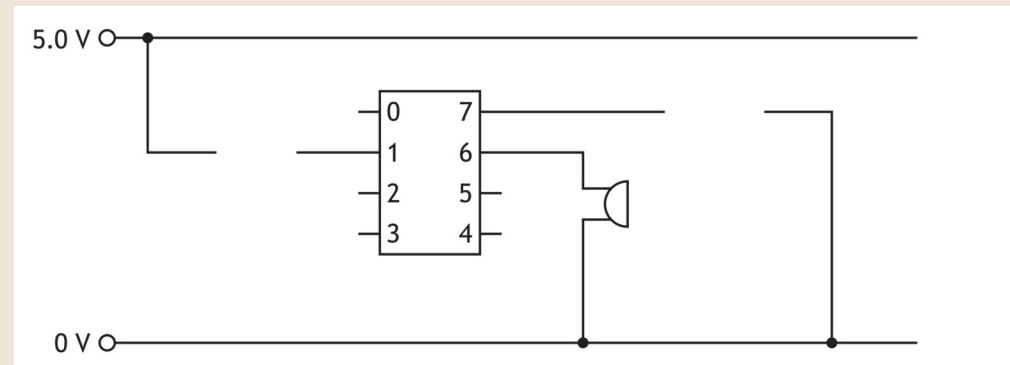
switch on pin 2

A pupil drew
this

Your job

find the mistake;
what does it cost?





2023 Q9(d) (2 marks)

2025 Q9(b) (2), (c) (1)

Past Papers: Answers



1 2023 Q9(d)

Sensor to pin 2 (1); pin
7 to base (1)

2 2023 Q9(d), also

No extra pins wired

3 2025 Q9(b)

Switch 5 V to pin 1 (1);
lamp pin 7 to 0 V (1)

4 2025 Q9(c)

Transistor (relay,
MOSFET; not 'driver')

Checkpoint



1 Where does an input switch connect?

From 5 V to the pin

2 Where does a buzzer connect?

From the pin to 0 V

3 Why can't a motor go straight on a pin?

Too little current — so
a transistor

Plenary: Part 2



1 An input switch goes from ... to the pin.

5 V

2 A small output goes from the pin to...

0 V

3 A motor needs what between it and the pin?

A transistor

4 What goes across the motor?

A protection diode

5 A wire to a pin not in the I/O table...

Loses the mark

Extension

EXTRA PRACTICE



1 Why is a diode put across the motor?

Protects the transistor from the spike

2 Why the resistor in the base lead?

It limits the base current

3 A 12 V lamp on pin 5: straight on, or not?

Through a transistor, with a 12 V supply

Look back — by the end of this part...



- 1 I can wire an input switch to a pin
- 2 I can wire an output to a pin
- 3 I can wire a transistor driver to a pin
- 4 I can complete a wiring diagram from an I/O table

LOOKING AHEAD

Lesson 2: Flowcharts

The wiring is done. How do we tell the chip what to do?

Course Specification

Digital electronic control systems

- examples of using microcontrollers in commercial and industrial applications
- advantages and disadvantages of microcontroller-based control systems, compared to a hard-wired electronic equivalent

Wiring to pins: asked in the 2023 and 2025 papers.

Credits (1 of 4)

The big question (A washing machine): Photo: Fanti Salms, CC BY-SA 3.0, via Wikimedia Commons

Title slide (Arduino Uno board): Photo: JrawX, CC0, via Wikimedia Commons

What Is a Microcontroller? (Arduino Uno board): Photo: JrawX, CC0, via Wikimedia Commons

Where You Meet Them (ticket machine): Photo: Infrastorian, CC0, via Wikimedia Commons

Where You Meet Them (packing line): Photo: SimónK, CC BY-SA 4.0, via Wikimedia Commons

Where You Meet Them (bus doors): Photo: Leif Jørgensen, CC BY-SA 4.0, via Wikimedia Commons

Where You Meet Them (smart lighting): Photo: Danielrosehill, CC BY 4.0, via Wikimedia Commons

Credits (2 of 4)

Advantages (marking-instruction wording): © Qualifications Scotland (SQA). N5 Engineering Science marking instructions: 2021 Q7(b), 2022 Q14(c), 2025 Q8(c).

Advantages (Arduino Uno board): Photo: JrawX, CC0, via Wikimedia Commons

Disadvantages (Arduino Uno board): Photo: JrawX, CC0, via Wikimedia Commons

Watch Out (marking instructions): © Qualifications Scotland (SQA). N5 Engineering Science marking instructions: 2021 Q7(b), 2022 Q14(c), 2025 Q8(c).

Model Answer: Describe (bus doors): Photo: Leif Jørgensen, CC BY-SA 4.0, via Wikimedia Commons

Model Answer: Explain (ticket machine): Photo: Infrastorian, CC0, via Wikimedia Commons

Spot the Mistake (marking instructions): © Qualifications Scotland (SQA). N5 Engineering Science marking instructions: 2021 Q7(b), 2022 Q14(c), 2025 Q8(c).

Credits (3 of 4)

A Motor Needs a Transistor (2025 Q9(c) marking instructions): © Qualifications Scotland (SQA). N5 Engineering Science 2023 Q9(d) and 2025 Q9(b)(c), question papers and marking instructions.

Model Answer: Complete the Wiring (2023 Q9(d) MI): © Qualifications Scotland (SQA). N5 Engineering Science 2023 Q9(d) and 2025 Q9(b)(c), question papers and marking instructions.

The Same Circuit in the Simulator: NoSyntaxSim V2 screenshot (hardware panel). Simulator by Peter Strain, NoStrainSims (sites.google.com/view/nostrainsim), free and open to all.

Spot the Mistake (2023 Q9(d) MI): © Qualifications Scotland (SQA). N5 Engineering Science 2023 Q9(d) and 2025 Q9(b)(c), question papers and marking instructions.

Credits (4 of 4)

Past Papers: Wiring: © Qualifications Scotland (SQA). N5 Engineering Science 2023 Q9(d) and 2025 Q9(b)(c), question papers and marking instructions.

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